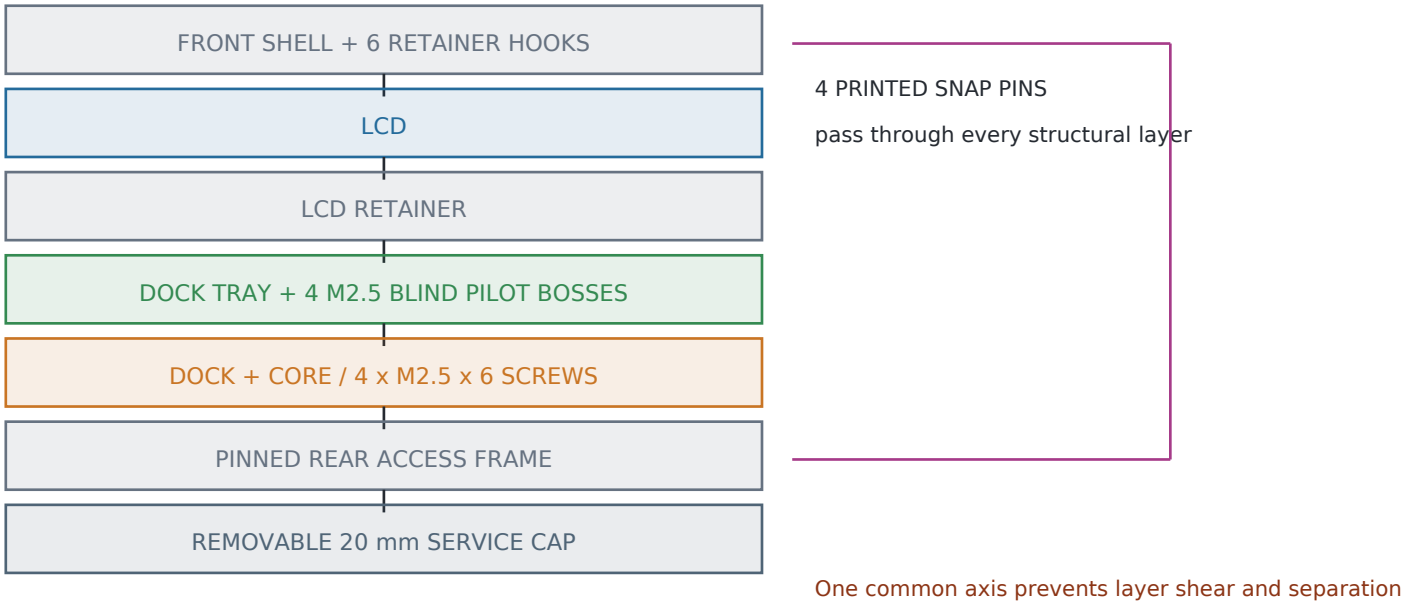
Tang Primer 20K
Integrated LCD Enclosure R4

Retention, assembly, tolerance and verification specification

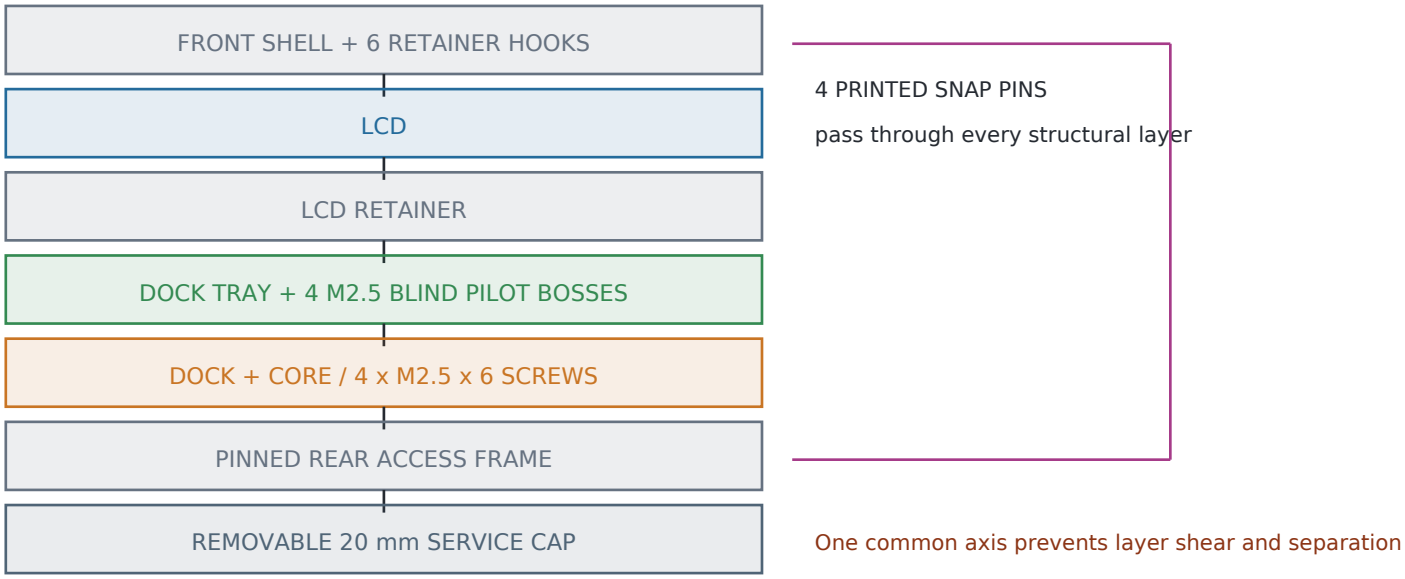


The LCD retainer, outer case and rear service cap are tool-less. The Dock uses its four official corner holes and M2.5 screws; no printed peg passes through the PCB.

PROTOTYPE STATUS: official Dock 3713 and Core 3690 STEP assemblies now define the installed SODIMM position and component envelope. Physical hardware must still verify PCB revision, FPC routing and FDM fit. Drop, vehicle vibration and safety certification are outside this revision.

1. Requirements and retention paths

No internal part may move freely during a light hand-shake test. The pinned access frame remains structural while the 20 mm service cap can be removed independently for SODIMM and PMOD access.



Retention summary

Target	Primary retention	Release
LCD	Outline pocket + edge retainer + six hooks	Deflect hooks outward and lift
Dock	4 x M2.5 x 6 screws through official holes	Remove screws; lift vertically
Core	SODIMM contacts + both connector latches	Open both latches together
FPC	Connector lock + open guide + polyimide tape	Remove tape; unlock connector
Case	Four printed split pins through 4.30 square paths	Squeeze rear tips and push forward
Service cap	Two upper hooks + two lower push tabs	Press lower tabs, rotate out, lift

2. Printed parts and assembly stack

STL	Qty	Function
front_shell_43_snap / 50_snap	1	LCD pocket, six hooks and front enclosure
lcd_retainer_43_snap / 50_snap	1	Loads LCD metal perimeter
dock_tray_screw_common	1	Dock support, blind pilots and open FPC guide
rear_access_frame_common	1	Pinned structure with 120 x 92 access opening
rear_service_cap_common	1	20 mm service bay, hooks, tabs and cable exits
case_snap_pin	4	Aligns and clamps all structural layers
fit_coupon	1	Case-pin and M2.5-pilot calibration

Stack order

Front to rear: front shell, LCD, LCD retainer, Dock tray, Dock with latched Core, pinned rear access frame, and removable service cap. Four printed pins stop at the frame; the cap has its own release tabs.

The access frame has a 120.00 x 92.00 central opening. Removing only the cap exposes the SODIMM latches and PMOD area without disturbing the LCD, tray, Dock screws or case pins.

Retainer hooks are placed two per side and two along the upper edge. The lower edge remains free for the 80 mm FPC route.

The Dock lands on four 7 x 7 x 3.2 support bosses. Four M2.5 x 6 pan-head screws pass through the official PCB holes and self-tap into 2.00 mm square, 4.80 mm deep blind pilots.

The 48 mm roof bridge in the former FPC tunnel has been removed. The release tray has two open rails and removable polyimide tape strain relief.

3. Dimensions and tolerances

Item	Dimension	Basis
Case	140.00 x 112.00 x 53.80	R4: Y +20, Z +20
Structural frame plane	Z=33.80	Original 37 mm pins retained
Service extension	20.00	Removable cap
Access opening	120.00 x 92.00	SODIMM / PMOD access
Dock	95.00 x 73.00	Official
Dock hole pitch	87.00 x 65.00	Official
Core PCB	67.60 x 30.00	Official
4.3 LCD	105.40 x 67.10 x 2.90	Official
5.0 LCD	120.70 x 75.90 x 3.05	Official
LCD pocket	LCD outline +0.50	0.25 per side
Case hole / pin	4.30 square / 3.50 square	0.40 per side
Dock PCB thickness	1.590 STEP	Official STEP
Dock boss / pilot	7.00 square / 2.00 square x 4.80 deep	Design
Dock screw	M2.5 x 6 pan-head, quantity 4	Hardware requirement
FPC guide	50.00 between outer rail faces; open top	Design - verify

R4 retains the 22.00 mm structural rear frame and 37.00 mm case pin, then adds a separate 20.00 mm service bay. Overall depth is 53.80 mm. Minimum STEP-envelope clearance to the service-cap inner face is 21.235 mm.

Fit adjustment

- Increase PIN_HOLE in 0.10 increments if insertion is too tight.

- Decrease PIN_HOLE in 0.10 increments if lateral play is excessive.

- Adjust DOCK_PILOT in 0.10 mm increments using the coupon; do not enlarge the PCB holes.

- Never reduce the LCD pocket to create clamping pressure on the display.

4. Printing and assembly

Part	Material	Layer	Walls / infill	Orientation
Case parts	PETG	0.20 max	4 / 20-30%	Largest flat face down
Service cap	PETG	0.16-0.20	4 / 25-35%	Rear lattice down
Dock tray	PETG	0.16 preferred	4 / 30%	Tray plane down
Snap pins	PETG, PA or PP	0.16	4 / 100%	As supplied: flat
Coupon	Production material	Production profile	Same	Do not rotate

Do not use PLA for repeated operation of retainer hooks or snap pins. Enable elephant-foot compensation. Determine horizontal-hole compensation from the coupon.

Assembly sequence

- Print the coupon and one case pin first; verify insertion and release.
- Place the LCD from the rear with FPC toward the lower edge.
- Insert the retainer from the upper edge; engage four side and two upper hooks.
- Lay the FPC between the open guide rails and add removable polyimide tape without tension.
- Place the Dock on all four bosses, then tighten four M2.5 x 6 screws evenly by hand.
- Insert the Core into SODIMM and close both side latches.
- Connect and lock the FPC without a sharp fold.
- Install the rear access frame and insert four case snap pins from the front.
- Insert both upper cap hooks, rotate the lower edge inward, and confirm both lower tabs click.

5. Disassembly and verification

Disassembly

- Disconnect power and all external cables.

- Press both lower service-cap tabs, rotate the lower edge outward, then lift off the upper hooks.

- For routine SODIMM or PMOD access, leave the four case pins, frame and Dock tray installed.

- For full teardown, squeeze each frame-side pin tip and push the pin toward the front.

- Unlock FPC and SODIMM latches before removing either board.

- Release side retainer hooks before the upper hooks. Never press the display area.

Acceptance checks

Stage	Check	Pass criterion
PDF	100 mm bar and 20 x 20 square	Printed dimensions match a ruler
Coupon	Case pin	Finger insertion; removable by squeezing tips
Coupon	M2.5 pilot	Screw holds without splitting or bottoming
Assembly	6 LCD hooks, 4 screws, 4 pins	All retainers seated; no board bending
Rear cap	2 hooks + 2 push tabs	Both detents click; finger release works
Hand shake	Three axes, 10 seconds each	No rattle or part movement
FPC	Connector and bend	No tension and no sharp crease
Powered	30 minutes	No heat deformation

6. Constraints and primary sources

Open verification items

- Official 2D drawings omit the SODIMM datum; the official Dock 3713 STEP supplies the installed Core position used here.

- Dock PCB is 1.590 mm in STEP; actual hole diameter and screw fit must be confirmed on hardware.

- Confirm FPC thickness, stiffener thickness and bend allowance for both LCD variants.

- Retention force depends on material, print orientation, moisture and extrusion accuracy.

- Drop, vehicle and long-duration vibration are untested. Use metal fasteners when required.

Primary sources

Sipeed Tang Primer 20K Wiki:
https://wiki.sipeed.com/hardware/en/tang/tang-primer-20k/primer-20k.html

Sipeed Dock/Core dimensions:
https://dl.sipeed.com/shareURL/TANG/Primer_20K/08_Dimensions

Sipeed LCD datasheets:
https://dl.sipeed.com/shareURL/TANG/Nano%209K/6_Chip_Manual/EN/LCD_Datasheet

DESIGN_R4.md is the Japanese companion specification. STL files, 1:1 drawings, manifest and tests shall be updated in the same commit whenever retention dimensions change.